

ADITYA VADALI

Dual Degree (B.Tech + MS by Research) Student, Computer Science
Robotics Research Center, IIIT Hyderabad, India
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EDUCATION

International Institute of Information Technology, Hyderabad Aug 2023 – May 2028 (expected)
B.Tech + MS by Research in Computer Science, Dual Degree (CSD) CGPA: 8.04 / 10

- Currently in 4th year (Semester 7).

PUBLICATIONS

[C1] Plann3r: Predicting Planning Costs Grounded in 3D

A. Vadali, K. Pandya, V. Garg, N. Chandna, D. Craggs, F. Dayoub, K. M. Krishna, S. Garg
Conference on Robot Learning (CoRL), 2026

First author. Accepted; presenting in person at CoRL 2026, Austin, TX, USA (Nov 9–12, 2026).

[C2] PixelLoop: Shortcut Topological Navigation with Pixel-Level Loops

S. Chittawar, V. Garg, A. Vadali, K. Pandya, R. Jayanti, S. Garg, K. M. Krishna
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2026
Accepted. [arXiv:2607.12811](https://arxiv.org/abs/2607.12811)

RESEARCH EXPERIENCE

Undergraduate Researcher, Robotics Research Center (RRC), IIIT Hyderabad May 2024 – Present
Advisor: Prof. K. Madhava Krishna · Collaborator: Prof. Sourav Garg

Plann3r: 3D-grounded planning costs

CoRL 2026, first author

- Fine-tuned VGGT with a new prediction head that outputs goal-conditioned geodesic costmaps for navigation from single-camera RGB input; evaluated on HM3D scenes.
- Designed and ran the rebuttal experiments, including a loop-closure extension addressing single-traversal assumptions, retrieval-noise sweeps, and bootstrap confidence intervals.

PixelLoop: pixel-level loop closures for topological navigation

IROS 2026

- Co-developed a loop-detection pipeline that adds pixel-level loop closures to topological navigation methods such as MAST3R-Nav, and ran its benchmark evaluation.

Ongoing work

- Topological navigation, visual place recognition (VPR), and RGB-only last-mile docking for mobile robots.

Deep-learning-based visual servoing

Summer 2024

- Revived legacy codebases for the lab's RTVS and Imagine2Servo methods and built a benchmark evaluating them across varying viewpoint angles and query-to-goal distances.

AWARDS & HACKATHONS

1st Place, Megathon 2025, IIIT Hyderabad (Qualcomm-sponsored)

Oct 2025

- **SnapGen**: prompt-driven Stable Diffusion style-transfer editor, deployed fully on-device on the Qualcomm QIDK NPU.

2nd Place, Megathon 2024, IIIT Hyderabad (Qualcomm-sponsored)

Oct 2024

- **Yoga instructor app**: on-device human keypoint detection on the Qualcomm QIDK NPU, with real-time pose-correction feedback from joint angles.

3rd Place, Lloyds Banking Group Hackathon

Dec 2025

- [Agentic AI system](#) on Google Cloud that detects and corrects data inconsistencies with a human in the loop.

Most Impactful Project, HackIIIT 2026, IIIT Hyderabad

Feb 2026

- Mess crowd-prediction system using live QR-scan data and ARMA forecasting to recommend visit times; integrated into the official MyIIIT app and used daily by 1000+ students.

SELECTED PROJECTS

Splatt3r-SfM | 3D Vision | [GitHub](#)

2025

- Combined MAST3R-SfM N-view pose estimation with Splatt3r's 2-view Gaussian head, fusing pairwise splats into a single aligned scene for feed-forward novel view synthesis from unposed images.

NoMaD Deployment on Pioneer P3DX | Robot Navigation | [GitHub](#)

2025

- Deployed an image-goal navigation model on a P3DX robot over ROS for closed-loop control, optimizing inference for real-time onboard operation.

Construction Site Safety Hazard Checker | Edge Computer Vision

- Fine-tuned YOLOv11 on an edge device to detect helmet, safety-vest and glove compliance, with an admin dashboard for worker-density monitoring and email hazard alerts.

Ray Tracer in C++ | Computer Graphics | [GitHub](#)

2024

- Full ray-tracing pipeline with analytic intersections for five primitive types, lighting, shading and shadows, rendered via OpenGL.

SKILLS

Languages Python, C++

Tools PyTorch, ROS, Docker, OpenGL

Research areas Visual navigation, 3D vision, visual place recognition, Gaussian splatting, robot planning